UNIVERSITY OF THE PEOPLE

MATH 1280-01 - AY2026-T4

Assignment Activity Unit 4

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Probability Distribution Analysis for FreshMart Groceries

Introduction

This assignment analyzes probability and statistical data for FreshMart Groceries, a small supermarket specializing in fresh produce and daily essentials. Using the tools of discrete probability distributions, expected value, standard deviation, and binomial probability, this report aims to provide data-driven insights to support inventory management, pricing strategies, and quality control. As Illowsky et al. (2023) explain, probability distribution functions are foundational tools that assign probabilities to outcomes and help describe patterns in real-world data, making them directly applicable to business decision-making scenarios such as those faced by FreshMart.

Question 1: Validating a Discrete Probability Distribution

Fresh Mart has recorded the purchase probabilities of four popular snack types based on historical sales data. To verify that these probabilities form a valid discrete probability distribution, two conditions must be satisfied (Illowsky et al., 2023, p. 228):

- Each probability must be between 0 and 1, inclusive.
- The sum of all probabilities must equal exactly 1.

The given purchase probabilities are shown in the table below:

Snack Type	Probability P(x)
Chips	0.35
Cookies	0.30
Nuts	0.20
Crackers	0.15

Verification of Condition 1: Each probability lies between 0 and 1:
P(Chips) = 0.35, P(Cookies) = 0.30, P(Nuts) = 0.20, P(Crackers) = 0.15. All values satisfy $0 \leq P(x) \leq 1$. ✓ Condition 1 is satisfied.

Verification of Condition 2: Sum of all probabilities equals 1:
P(Chips) + P(Cookies) + P(Nuts) + P(Crackers) = 0.35 + 0.30 + 0.20 + 0.15 = 1.00
✓ Condition 2 is satisfied.

Since both conditions are met, the given data forms a valid discrete probability distribution. This means the model accurately accounts for all possible snack purchase outcomes, and each outcome has been assigned a meaningful and complete probability.

If the probabilities did not sum to 1, the distribution would be invalid because it would imply either that some outcomes are missing from the model, or that the recorded frequencies contain errors. A sum greater than 1 would suggest over-counting, while a sum less than 1 would indicate unaccounted outcomes. According to Illowsky et al. (2023), any distribution that fails these two conditions cannot be considered a valid probability distribution function.

Question 2: Computing Expected Value and Standard Deviation

FreshMart sells fresh juice in three sizes at different prices with associated customer purchase probabilities. The random variable X represents the revenue per juice purchase.

Juice Size	Price (x)	P(x)	$x \cdot P(x)$	$x^2 \cdot P(x)$
Small	\$3.50	0.40	$3.50 \times 0.40 = 1.400$	$12.25 \times 0.40 = 4.900$
Medium	\$5.00	0.35	$5.00 \times 0.35 = 1.750$	$25.00 \times 0.35 = 8.750$
Large	\$7.00	0.25	$7.00 \times 0.25 = 1.750$	$49.00 \times 0.25 = 12.250$
Total	—	1.00	$\mu = 4.900$	$\Sigma x^2P(x) = 25.900$

Step 1 – Expected Value (Mean):

The expected value formula is: $\mu = \Sigma[x \cdot P(x)]$ (Illowsky et al., 2023, p. 231)

$\mu = (3.50)(0.40) + (5.00)(0.35) + (7.00)(0.25)$

$\mu = 1.40 + 1.75 + 1.75$

$\mu = \$4.90$

Interpretation: On average, FreshMart earns \$4.90 per juice purchase. Over thousands of transactions, the long-term revenue per sale will converge to this value, consistent with the Law of Large Numbers (Illowsky et al., 2023, p. 230).

Step 2 – Standard Deviation:

The standard deviation is computed using the variance formula: $\sigma^2 = \Sigma[x^2 \cdot P(x)] - \mu^2$ (Illowsky et al., 2023, p. 232)

$\Sigma[x^2 \cdot P(x)] = 4.900 + 8.750 + 12.250 = 25.900$

$\mu^2 = (4.90)^2 = 24.010$

$\sigma^2 = 25.900 - 24.010 = 1.890$

$\sigma = \sqrt{1.890} \approx \$1.374$

Interpretation: The standard deviation of approximately $1.37 indicates a moderate variation in revenue per juice sale. This means individual transactions may deviate from the $4.90 average by roughly $1.37. For FreshMart, this level of variability is manageable and supports consistent pricing strategy planning.

Question 3: Applying the Binomial Distribution

FreshMart's bakery division tests loaves of bread with a 90% success rate (p = 0.90). Each loaf is tested independently, and the outcome is binary (pass/fail), satisfying the three conditions of a binomial experiment: fixed number of trials, only two possible outcomes, and independent trials with constant probability (Illowsky et al., 2023, p. 237). The random variable X represents the number of loaves meeting the quality standard.

The binomial probability formula is:

$$P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{(n-k)}$$

Part (a): P(X = 10) with n = 12, p = 0.90

Given: n = 12, k = 10, p = 0.90, q = 0.10

$P(X = 10) = C(12, 10) \cdot (0.90)^{10} \cdot (0.10)^2$

Step 1 – Combination:

$C(12, 10) = 12! / (10! \times 2!) = (12 \times 11) / (2 \times 1) = 66$

Step 2 – Powers:

$(0.90)^{10} \approx 0.34868$

$(0.10)^2 = 0.01$

Step 3 – Final Calculation:

$P(X = 10) = 66 \times 0.34868 \times 0.01$

$P(X = 10) = 66 \times 0.0034868$

$P(X = 10) \approx 0.2301$

Interpretation: There is approximately a 23.01% probability that exactly 10 out of 12 tested loaves will pass quality control. This information can help FreshMart anticipate daily defect rates and plan for appropriate staffing or re-testing procedures.

Part (b): P(X ≥ 8) with n = 10, p = 0.90

Given: $n = 10$, $p = 0.90$, $q = 0.10$

$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10)$

Calculating $P(X = 8)$:

$C(10, 8) = 10! / (8! \times 2!) = 45$

$(0.90)^8 \approx 0.43047$

$(0.10)^2 = 0.01$

$P(X = 8) = 45 \times 0.43047 \times 0.01 \approx 0.1937$

Calculating $P(X = 9)$:

$C(10, 9) = 10$

$(0.90)^9 \approx 0.38742$

$(0.10)^1 = 0.10$

$P(X = 9) = 10 \times 0.38742 \times 0.10 \approx 0.3874$

Calculating $P(X = 10)$:

$C(10, 10) = 1$

$(0.90)^{10} \approx 0.34868$

$(0.10)^0 = 1$

$P(X = 10) = 1 \times 0.34868 \times 1 \approx 0.3487$

Final Sum:
$P(X \geq 8) = 0.1937 + 0.3874 + 0.3487 \approx 0.9298$

Interpretation: There is approximately a 92.98% probability that at least 8 out of 10 loaves will meet the quality standard. This high probability reflects FreshMart's strong quality control process and gives management confidence in the consistency of their bakery production. Such analysis is critical for maintaining customer trust and reducing waste (Illowsky et al., 2023, p. 241).

Conclusion

This analysis demonstrates the practical power of probability and statistics in a real-world retail context. The snack purchase data was confirmed as a valid discrete probability distribution. The expected revenue per juice sale was calculated at $4.90 with a standard deviation of approximately $1.37, enabling FreshMart to plan pricing and inventory with greater accuracy. Finally, the binomial distribution revealed that the bakery maintains a high quality control success rate, with a 92.98% probability that at least 8 of every 10 loaves meet standards. Together, these tools allow FreshMart to make informed, data-driven operational decisions.

References

Illowsky, B., Dean, S., Birmajer, D., Blount, B., Boyd, S., Einsohn, M., Helmreich, J., Kenyon, L., Lee, S., & Taub, J. (2023). *Introductory statistics 2e.* OpenStax. https://openstax.org/details/books/introductory-statistics-2e/